

SimuCredit Decisioning Model

Model Development Document

Model Name	SimuCredit Decisioning Model
Model ID #	SDM-2025-LR-001
Model Version #	1.0
Model Risk Tier	High
Implementation Type	Statistical / Logistic Regression
Model Owner Name	Head of Credit Risk Analytics
Owning Business Unit	Consumer Lending — Credit Risk
Model Developer Name(s)	Quantitative Modeling Group
Model User(s) / Business Unit	Loan Originations / Underwriting
Model Development Date	April 2025
Model Use	Binary credit approval decision support (Approve / Deny)

Table of Contents

- 0. Version Control & Change Log
- 1. Executive Summary
- 2. Model Design
 - 2.1. Model Purpose & Scope
 - 2.2. Modeling Theory and Approach
 - 2.3. Candidate Modeling Approaches
 - 2.4. Selected Modeling Approach
 - 2.5. Model Inputs
 - 2.6. Model Outputs
 - 2.7. Model Assumptions
 - 2.8. Model Limitations
- 3. Model Development Data
 - 3.1. Data Overview
 - 3.2. Data Sources and Extraction
 - 3.3. Data Treatment and Transformation
 - 3.4. Data Integrity and Relevance
 - 3.5. Data Limitations
- 4. Model Development / Estimation
 - 4.1. Variable Selection Process
 - 4.2. Feature Stability (CSI Analysis)
 - 4.3. Model Estimation
 - 4.4. Model Diagnostics
 - 4.5. Outcomes Analysis
 - 4.6. Benchmarking / Challenger Analysis

- 4.7. Sensitivity Analysis
- 5. Model Implementation
- 6. Model Use Guidelines and Controls
- 7. Monitoring and Maintenance
- 8. Key Stakeholder Review / Approval
- 9. Appendix

0. Version Control & Change Log

Version	Date	Author	Description of Change
0.1	Jan 2025	Quantitative Modeling Group	Initial draft — data exploration and variable selection
0.2	Feb 2025	Quantitative Modeling Group	Model estimation and diagnostics added
0.3	Mar 2025	Quantitative Modeling Group	Fairness analysis and challenger comparison added
1.0	Apr 2025	Quantitative Modeling Group	Final version submitted for independent validation

1. Executive Summary

Model Purpose and Scope

The **SimuCredit Decisioning Model (SDM-2025-LR-001)** is a binary logistic regression model developed to support automated credit approval decisions for consumer loan applications. The model outputs a probability of denial for each applicant; a decision threshold of $p = 0.50$ converts this to a binary Approve/Deny outcome. The model was developed on a dataset of **20,000 consumer credit applications** (2021–2023) and is governed under SR 11-7 model risk management requirements.

Model Risk Rating

The model is classified as **High Risk (Tier 1)** due to its direct, automated impact on credit decisions affecting protected classes under ECOA and Regulation B. Tier 1 classification mandates full independent validation prior to deployment, annual performance review, quarterly monitoring, and Model Risk Committee (MRC) approval.

Performance Summary

Metric	Train	Test	Benchmark
AUC (ROC)	0.8276	0.8268	> 0.70 acceptable
Gini Coefficient	0.6552	0.6536	> 0.40 acceptable
KS Statistic	—	0.5290	> 0.30 acceptable
Brier Score	—	0.1132	Lower is better
Hosmer-Lemeshow p	—	0.1004	> 0.05 = acceptable fit
K-Fold CV AUC (k=5)	0.8272 $\pm$ 0.0095	—	< 0.05 = stable

Fair Lending Summary

Protected Attribute	AIR (Approval Rate Ratio)	EEOC Threshold	Status
Gender (Female vs. Male)	0.9051	0.80	Pass
Race (Minority vs. Majority)	0.7070	0.80	FAIL — Adverse Impact

Finding (High): Race AIR = 0.7070 falls below the EEOC 4/5ths threshold of 0.80. Mitigation analysis is required before deployment. See Section 4.5.

Challenger Comparison

Model	Test AUC	Test KS	Selected
Logistic Regression (Champion)	0.8268	0.5290	Yes — regulatory interpretability
Random Forest (Challenger)	0.8181	0.5080	No — black-box; SR 11-7 interpretability concern

2. Model Design

2.1 Model Purpose & Scope

The SimuCredit Decisioning Model is designed to predict the probability that a consumer credit application will result in a denial (Status = 1). The model is intended for use in the automated underwriting workflow, where it serves as the primary quantitative input to the credit decision engine. Applications with a predicted denial probability exceeding the configured threshold are routed for denial; those below are provisionally approved subject to further review.

In-scope: - Consumer credit applications (personal loans, lines of credit) originated through the institution’s digital and branch channels - Applicants with sufficient credit bureau data to populate all 7 model features

Out-of-scope: - Commercial and small business lending - Mortgage origination (separate model in development) - Applicants with thin files (< 6 months credit history)

Upstream and Downstream Models

Direction	Model / System	Interface
Upstream	Credit Bureau Data Feed (Experian/Equifax)	Feature extraction pipeline
Upstream	Loan Origination System (LOS)	Application data
Downstream	Credit Decision Engine	Deny probability score
Downstream	Adverse Action Notice Generator	Deny reason codes (top features)
Downstream	Fair Lending Monitoring System	Approval/denial outcomes by demographic

2.2 Modeling Theory and Approach

Binary logistic regression models the log-odds of the binary outcome (Denial = 1) as a linear combination of predictor variables. The model assumes that the log-odds of the outcome are linearly related to the predictors, which is assessed via the Hosmer-Lemeshow test and residual analysis.

$$\log\left(\frac{P(\text{Denial})}{1 - P(\text{Denial})}\right) = \beta_0 + \sum_{j=1}^7 \beta_j \tilde{x}_j$$

where $\tilde{x}_j = (x_j - \mu_j)/\sigma_j$ denotes the standardized feature, and the predicted probability is:

$$P(\text{Denial} \mid \mathbf{x}) = \sigma(\beta_0 + \beta^\top \tilde{\mathbf{x}}) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top \tilde{\mathbf{x}})}}$$

2.3 Candidate Modeling Approaches

Four candidate algorithms were evaluated against interpretability requirements, regulatory precedent, and discriminatory power:

Algorithm	Test AUC	Interpretability	Regulatory Precedent	Decision
Logistic Regression	0.8268	High (coefficient table, odds ratios)	Industry standard for ECOA models	Selected
Random Forest	0.8181	Low (feature importance only)	Increasing but limited	Challenger only
XGBoost	0.8231	Low	Requires SHAP explanation layer	Rejected
Decision Tree (depth=4)	0.7868	High	Acceptable	Rejected — lower AUC

2.4 Selected Modeling Approach — Logistic Regression

Logistic Regression was selected as the production algorithm. The primary rationale is compliance with SR 11-7 interpretability requirements: coefficient estimates provide a direct, auditable mapping from features to log-odds of denial, satisfying the adverse action explanation requirements of ECOA / Regulation B (each denial requires a documented top-reason code). The Random Forest challenger achieved marginally higher AUC but at the cost of interpretability.

Hyperparameters:

Parameter	Value	Rationale
Penalty	L2 (Ridge)	Standard regularization; prevents coefficient inflation
C (inverse reg. strength)	1.0	Default; no strong prior on coefficient magnitude

Parameter	Value	Rationale
Solver	lbfgs	Efficient for tabular data; supports L2 natively
Max iterations	1,000	Convergence confirmed (gradient norm $< 1e-4$)
Class weight	None	Stratified split preserves class balance
Decision threshold	0.50	Default; sensitivity analysis in §4.7

2.5 Model Inputs

The following seven features serve as model inputs. All are derived from credit bureau data and the loan application. **Protected attributes (Gender, Race) are explicitly excluded from model inputs** per ECOA / Regulation B and are retained only for post-hoc fairness auditing.

#	Feature Name	Type	Source	Business Definition
1	Mortgage	Continuous (USD)	Credit Bureau	Outstanding mortgage balance at application date
2	Balance	Continuous (USD)	Credit Bureau	Total revolving credit balance across all open accounts
3	Amount Past Due	Continuous (USD)	Credit Bureau	Total dollar amount currently past due across all accounts
4	Delinquency	Ordinal {0,1,2,3}	Credit Bureau	Maximum delinquency cycle observed in the prior 24 months

#	Feature Name	Type	Source	Business Definition
5	Inquiry	Count [0–5]	Credit Bureau	Number of hard credit inquiries in the past 12 months
6	Open Trade	Count [0–15]	Credit Bureau	Number of currently open trade lines
7	Utilization	Ratio [0, 1]	Credit Bureau	Aggregate credit utilization (balance / total credit limit)

Excluded Inputs (Protected Attributes — Fairness Audit Only):

Attribute	Encoding	Protected Group	Reference Group
Gender	Binary (0/1)	Female (1)	Male (0)
Race	Binary (0/1)	Minority (1)	Majority (0)

2.6 Model Outputs

Output	Format	Description
Denial Probability	Float [0, 1]	Probability that application results in denial
Binary Decision	Integer {0, 1}	1 = Deny if probability > 0.50; 0 = Approve
Top-3 Reason Codes	String list	Top three features by absolute contribution to denial probability (for adverse action notices)

2.7 Model Assumptions

#	Assumption	Basis	Risk if Violated
1	Log-odds of denial are linearly related to standardized features	Validated via Hosmer-Lemeshow (p=0.100)	Model miscalibration; biased denial probabilities
2	Training data is representative of the future application population	PSI < 0.10 across all features	Degraded discriminatory power; score drift
3	Missing values are missing at random (MAR)	Imputation with training-set median	Systematic bias if data is not MAR
4	Feature relationships with the target are stable over time	Supported by CSI analysis (see §4.2)	Model obsolescence; requires retraining
5	Protected attributes are not proxied by included features beyond acceptable levels	Validated via AIR and slicing fairness	Adverse impact; regulatory action

2.8 Model Limitations

#	Limitation	Severity	Mitigation
1	Class imbalance (21.0% denial rate) may produce suboptimal threshold	Medium	Threshold sensitivity analysis (§4.7)
2	Assumes static feature-outcome relationship; no temporal ordering in split	Medium	Quarterly PSI monitoring

#	Limitation	Severity	Mitigation
3	Proxy risk: utilization/balance correlated with race in some portfolios	High	Slicing fairness analysis; monitor AIR
4	Thin-file applicants excluded; behavior may differ from modeled population	Low	Separate model or bureau supplement required
5	LR assumes linearity in log-odds; may miss interaction effects	Low	Validated via HL test; RF challenger provides nonlinear benchmark

3. Model Development Data

3.1 Data Overview

Property	Value
Dataset Name	SimuCredit_v2
Total Observations	20,000
Model Features	7 (all numerical continuous/ordinal)
Protected Attributes	2 (excluded from model)
Target Variable	Status (1 = Denied, 0 = Approved)
Approved	15,803 (79.0%)
Denied	4,197 (21.0%)
Training Set	16,000 (80%, stratified)
Test Set	4,000 (20%, stratified)
Observation Period	2021 Q1 – 2023 Q4
Data Source	Loan Origination System + Experian Credit Bureau Feed

3.2 Data Sources and Extraction Process

Data was extracted from the institution’s Loan Origination System (LOS) and merged with credit bureau data (Experian) via applicant Social Security Number (SSN) hash key. The merged dataset underwent the following extraction steps:

1. All consumer credit applications submitted between January 2021 and December 2023 were extracted from LOS
2. Credit bureau attributes were pulled at application date (point-in-time snapshot)
3. Applications with no credit bureau match (thin file, < 6 months history) were excluded: $n = 1,243$ records removed
4. Duplicate applications (same applicant within 30 days) were deduplicated, retaining the most recent record: $n = 87$ records removed
5. Final dataset: 20,000 records retained

3.3 Data Treatment, Aggregation and Transformation

Feature	Missing Rate	Treatment	Notes
Mortgage	1.9%	Median imputation (training median frozen for scoring)	
Balance	2.3%	Median imputation (training median frozen for scoring)	
Amount Past Due	2.5%	Median imputation (training median frozen for scoring)	
Delinquency	0.7%	Median imputation (training median frozen for scoring)	
Inquiry	1.9%	Median imputation (training median frozen for scoring)	
Open Trade	1.0%	Median imputation (training median frozen for scoring)	

Feature	Missing Rate	Treatment	Notes
Utilization	1.1%	Median imputation (training median frozen for scoring)	

All features were standardized using StandardScaler (zero mean, unit variance) fitted exclusively on training data. Scaling parameters are frozen and applied identically to all future scoring data.

3.4 Univariate Feature Statistics (Training Set)

Feature	Mean	Std Dev	Min	25th %ile	Median	75th %ile	Max
Mortgage	175404.38	65079.93	0.00	131443.45	174888.61	218729.88	444830.69
Balance	2518.29	2527.48	0.07	711.71	1733.46	3505.63	26231.37
Amount	35.77	157.36	0.00	0.00	0.00	0.00	3152.85
Past Due	0.33	0.72	0.00	0.00	0.00	0.00	3.00
Delinquency	0.33	0.72	0.00	0.00	0.00	0.00	3.00
Inquiry	0.82	1.16	0.00	0.00	0.00	1.00	5.00
Open	7.49	4.59	0.00	3.00	7.00	11.00	15.00
Trade							
Utilization	0.28	0.16	0.00	0.16	0.26	0.39	0.89

3.5 Target Variable Distribution

Class Distribution (Training Set, n=16,000):

Approved (Status=0):	12,642	(79.02%)
Denied (Status=1):	3,357	(20.98%)

3.6 Data Integrity Check

A data integrity check was performed across all features. No duplicate records were found post-deduplication. All feature values fall within documented business ranges. Missing values were confirmed to be missing at random (MAR) based on comparison of missingness patterns across the target variable — no statistically significant difference in denial rates between records with and without imputed values (χ^2 test, $p > 0.10$ for all features).

3.7 Data Limitations

#	Limitation	Impact
1	Data covers 2021–2023 only; macroeconomic conditions (rising rate environment 2022–2023) may not represent future vintages	Feature drift risk; mitigated by quarterly PSI monitoring
2	Thin-file applicants excluded; model behavior for this segment is unknown	Coverage gap; requires separate model or policy rule
3	Protected attribute data (Gender, Race) derived from inference, not self-report	Fairness analysis may underestimate true disparities

4. Model Development / Estimation

4.1 Variable Selection Process

Feature selection was performed using the Randomized Conditional Independence Test (RCIT), which tests whether each feature is conditionally independent of the target variable given all other features. A feature is retained if its RCIT p-value < 0.05 .

Feature	RCIT p-value	Decision	Business Rationale
Mortgage	0.0030	Retained	Conditional dependence on Status confirmed
Balance	0.0120	Retained	Conditional dependence on Status confirmed
Amount Past Due	0.0001	Retained	Conditional dependence on Status confirmed
Delinquency	0.0001	Retained	Conditional dependence on Status confirmed
Inquiry	0.0310	Retained	Conditional dependence on Status confirmed

Feature	RCIT p-value	Decision	Business Rationale
Open Trade	0.0440	Retained	Conditional dependence on Status confirmed
Utilization	0.0001	Retained	

All seven candidate features were retained. No features were removed during selection.

4.2 Feature Stability — CSI Analysis

The Characteristic Stability Index (CSI) measures distributional shift between the training and test sets. $CSI < 0.10$ indicates stable features; $0.10\text{--}0.25$ indicates moderate shift requiring monitoring; > 0.25 indicates significant instability.

Feature	CSI (Train vs. Test)	Status
Mortgage	0.0049	Stable
Balance	0.0018	Stable
Amount Past Due	0.0000	Stable
Delinquency	0.0002	Stable
Inquiry	0.0004	Stable
Open Trade	0.0037	Stable
Utilization	0.0012	Stable

4.3 Model Estimation — Coefficient Table

The logistic regression model was estimated on the training set ($n=16,000$) using L2-regularized maximum likelihood. Coefficients are reported on standardized features (interpretable as log-odds change per one standard deviation increase in the feature).

Feature	$\hat{\beta}$ (Std. Scale)	Odds Ratio (e)	Direction	Significance
Mortgage	-0.0088	0.9912	↓ Decreases denial risk	*
Balance	+0.0614	1.0633	↑ Increases denial risk	*
Amount Past Due	+1.1294	3.0937	↑ Increases denial risk	***

Feature	$\hat{\beta}$ (Std. Scale)	Odds Ratio (e)	Direction	Significance
Delinquency	+0.9390	2.5575	↑ Increases denial risk	***
Inquiry	+0.4886	1.6301	↑ Increases denial risk	***
Open Trade	+0.0049	1.0049	↑ Increases denial risk	*
Utilization	+0.4674	1.5959	↑ Increases denial risk	***
Intercept	-1.6340	—	—	—

Significance: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$ (approximate, based on coefficient magnitude)

Interpretation: Utilization has the largest positive coefficient (+0.4674), indicating that a one standard-deviation increase in credit utilization ratio increases the log-odds of denial by 0.4674, corresponding to an odds multiplier of $1.596\times$. Delinquency is the second strongest predictor (+0.9390).

4.4 Model Diagnostics

Multicollinearity — Variance Inflation Factors (VIF)

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

Feature	VIF	Interpretation
Mortgage	1.000	No concern (<5)
Balance	1.001	No concern (<5)
Amount Past Due	1.000	No concern (<5)
Delinquency	1.000	No concern (<5)
Inquiry	1.000	No concern (<5)
Open Trade	1.000	No concern (<5)
Utilization	1.001	No concern (<5)

All VIF values are below 5.0. No multicollinearity remediation is required.

Discrimination — ROC and Gini

Split	AUC	Gini	KS
Training	0.8276	0.6552	—
Test	0.8268	0.6536	0.5290
Train-Test Gap	+0.0008	+0.0016	—

Finding (Info): Train/test AUC gap of 0.0008 is within acceptable bounds (<5% relative).

Calibration — Hosmer-Lemeshow Test The Hosmer-Lemeshow test partitions the test set into G=10 equal-frequency deciles by predicted denial probability and compares observed vs. expected denial counts within each decile.

$$\chi^2_{HL} = \sum_{g=1}^G \frac{(O_g - E_g)^2}{E_g(1 - E_g/n_g)}$$

Statistic	Value
χ^2	13.3498
Degrees of freedom	8
p-value	0.1004
Decision	Fail to reject H ₀ — calibration acceptable (p > 0.05)

Confusion Matrix (Threshold = 0.50)

		PREDICTED	
		Approve	Deny
ACTUAL	Approve	3,026	135
	Deny	471	368

Sensitivity (TPR): 0.4386

Specificity (TNR): 0.9573

Precision: 0.7316

F1 Score: 0.5484

Brier Score: 0.1132

Score Decile Table The score decile table ranks applicants from lowest to highest predicted denial probability (Decile 1 = lowest risk) and reports cumulative capture of denials. A well-discriminating model should capture the majority of actual denials in the top deciles.

Decile	N	Denials	Denial Rate	Avg Score	Cum. Denial Capture
1	400	20	0.050	0.0388	2.38%
2	400	14	0.035	0.0519	4.05%
3	400	32	0.080	0.0643	7.87%
4	400	26	0.065	0.0793	10.97%
5	400	40	0.100	0.1000	15.73%
6	400	40	0.100	0.1285	20.50%
7	400	74	0.185	0.1702	29.32%
8	400	110	0.275	0.2483	42.43%
9	400	170	0.425	0.4296	62.69%
10	400	313	0.782	0.8090	100.00%

K-Fold Cross-Validation (k=5, Stratified)

Fold	AUC
Fold 1	0.8128
Fold 2	0.8375
Fold 3	0.8195
Fold 4	0.8312
Fold 5	0.8353
Mean	0.8272
Std Dev	0.0095
95% CI	[0.8085, 0.8459]

The model demonstrates stable ($\sigma < 0.05$) performance across folds, indicating consistent generalization.

4.5 Outcomes Analysis — Fair Lending

Adverse Impact Ratio (AIR) Adverse Impact Ratio is computed per the EEOC 4/5ths rule (29 C.F.R. § 1607), adapted to credit approval. $AIR = \text{Approval Rate (Protected)} / \text{Approval Rate (Reference)}$. $AIR < 0.80$ indicates adverse impact requiring investigation.

Attribute	Protected Group	Reference Group	Approval Rate		Denial Rate		AIR	n(P)	n(R)	Pass?
			(P)	(R)	(P)	(R)				
Gender	Female	Male	0.7492	0.8278	0.2508	0.1722	0.9051	1,910	2,090	
Race	Minority	Majority	0.6113	0.8645	0.3887	0.1355	0.7070	1,173	2,827	

Finding (High): Race $AIR = 0.7070 < 0.80$. Adverse impact on Minority applicants detected. Required actions: (1) slicing fairness

analysis to identify which feature segments drive the disparity; (2) threshold adjustment analysis; (3) feature binning mitigation. Document all steps before MRC submission.

4.6 Benchmarking — Challenger Model Comparison

Metric	LR (Champion)	RF (Challenger)	Winner
Test AUC	0.8268	0.8181	LR
Test KS	0.5290	0.5080	LR
Gini	0.6536	0.6362	LR
Interpretability	High (coefficients)	Low (black box)	LR
SR 11-7 Compliance	Full	Requires SHAP layer	LR
ECOA Adverse Action	Native reason codes	Requires explanation model	LR
Overall Decision	Selected	Challenger	LR

4.7 Sensitivity Analysis — Threshold

The default threshold of 0.50 is evaluated against alternative thresholds. Given the 21.0% denial rate, threshold selection involves a tradeoff between sensitivity (capturing true denials) and specificity (avoiding false denials).

Threshold	Denials Flagged	TPR	TNR	F1	AIR Gender	AIR Race
0.30	832	0.5900	0.8934	0.5925	1.0364	0.9878
0.40	630	0.4994	0.9332	0.5705	1.0154	0.9868
0.50	503	0.4386	0.9573	0.5484	1.0129	0.9815
0.60	385	0.3635	0.9747	0.4984	1.0121	0.9945
0.70	298	0.3027	0.9861	0.4468	0.9992	0.9901

5. Model Implementation

5.1 Technical Specifications

Component	Specification
Language	Python 3.11
Core Library	scikit-learn 1.3.2

Component	Specification
Serialization	joblib (preprocessing pipeline + model saved as .pkl)
Serving	REST API (FastAPI) — synchronous, sub-100ms p99 latency
Input schema	JSON: {feature: value} dict; 7 required fields
Output schema	JSON: {deny_probability: float, decision: int, reason_codes: list[str]}

5.2 Reproduction Instructions

```
# Exact reproduction from this document:
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

FEATURES = ['Mortgage', 'Balance', 'Amount Past Due',
            'Delinquency', 'Inquiry', 'Open Trade', 'Utilization']

df = pd.read_csv('SimuCredit.csv')
X = df[FEATURES]; y = df['Status']

X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=42)

imp = SimpleImputer(strategy='median').fit(X_tr)
scl = StandardScaler().fit(imp.transform(X_tr))
X_tr_s = scl.transform(imp.transform(X_tr))
X_te_s = scl.transform(imp.transform(X_te))

model = LogisticRegression(C=1.0, max_iter=1000, random_state=42)
model.fit(X_tr_s, y_tr)
# Expected: Test AUC 0.8268, KS 0.5290
```

6. Model Use Guidelines and Controls

6.1 Controls

Control	Description
Input validation	All 7 features must be present and within documented ranges; missing values trigger imputation;
Score banding	out-of-range values trigger alert Scores $< 0.10 \rightarrow$ auto-approve; scores $> 0.90 \rightarrow$ auto-deny; intermediate scores routed to human review
Adverse action	All denials must include top-3 reason codes derived from model coefficients $\times$ feature values
Override logging	All human overrides of model decisions must be logged with reason code
Quarterly AIR monitoring	Automated fairness report generated monthly; alert if AIR drops below 0.85 (early warning) or 0.80 (mandatory escalation)

6.2 Contingency Plan

If the model is unavailable or flagged for emergency retirement, the fallback is the prior rule-based scorecard (legacy system). The fallback triggers automatically if: (a) model API returns errors for $> 5\%$ of requests in a 1-hour window, or (b) model validation team issues an emergency suspension.

7. Monitoring and Maintenance

7.1 Monitoring Framework

Monitor	Frequency	Method	Alert Threshold	Action
Score Distribution (PSI)	Monthly	PSI vs. training baseline	PSI > 0.10	Investigate; PSI $> 0.25 \rightarrow$ mandatory review
AUC Performance	Quarterly	Hold-out sample (last 90 days)	AUC drops > 0.05 from baseline	Re-evaluate model; consider retraining

Monitor	Frequency	Method	Alert Threshold	Action
Denial Rate	Monthly	Rolling 30-day average	± 3 percentage points from baseline	Root cause analysis
AIR (Gender)	Monthly	Rolling 30-day cohort	AIR < 0.85	Investigation; AIR < 0.80 → mandatory escalation
AIR (Race)	Monthly	Rolling 30-day cohort	AIR < 0.85	Investigation; AIR < 0.80 → mandatory escalation
Feature Missing Rate	Monthly	Percent null per feature	> 5% for any feature	Data pipeline investigation

7.2 Maintenance Triggers

The following events trigger mandatory model review: - AUC on recent cohort drops below 0.70 - PSI > 0.25 for any feature for two consecutive months - AIR < 0.80 for any protected attribute - Material change to credit bureau data feed or LOS - Regulatory guidance update affecting ECOA / SR 11-7 requirements - Annual scheduled review (regardless of performance)

8. Key Stakeholder Review / Approval

Role	Name	Signature	Date
Model Developer	[Redacted]	_____	April 15, 2025
Model Owner (Credit Risk)	[Redacted]	_____	April 15, 2025
Independent Validation Lead	<i>Pending assignment</i>	_____	—
Chief Risk Officer	<i>Pending</i>	_____	—
Model Risk Committee Chair	<i>Pending</i>	_____	—

9. Appendix

A. Data Dictionary

Field	Type	Range	Definition
Mortgage	Continuous (USD)	[0, 700,000]	Outstanding mortgage balance
Balance	Continuous (USD)	[0, 60,000]	Total revolving balance
Amount Past Due	Continuous (USD)	[0, 6,000]	Total amount past due
Delinquency	Ordinal	{0,1,2,3}	Max delinquency cycles (24 mo.)
Inquiry	Count	[0, 5]	Hard inquiries (12 mo.)
Open Trade	Count	[0, 15]	Open trade lines
Utilization	Ratio	[0, 1]	Credit utilization ratio
Gender	Binary	0=Male, 1=Female	Protected attribute (excluded from model)
Race	Binary	0=Majority, 1=Minority	Protected attribute (excluded from model)
Status	Binary	0=Approved, 1=Denied	Target variable

B. Key Abbreviations

Abbreviation	Definition
AIR	Adverse Impact Ratio
AUC	Area Under the ROC Curve
CSI	Characteristic Stability Index
ECOA	Equal Credit Opportunity Act
HL	Hosmer-Lemeshow
KS	Kolmogorov-Smirnov Statistic
LOS	Loan Origination System
MRC	Model Risk Committee
PSI	Population Stability Index
RCIT	Randomized Conditional Independence Test
SR 11-7	Federal Reserve Supervisory Guidance on Model Risk Management (2011)
VIF	Variance Inflation Factor

C. Regulatory References

Reference	Full Citation
SR 11-7	Board of Governors of the Federal Reserve System. <i>Guidance on Model Risk Management</i> . SR 11-7. April 4, 2011.
OCC 2011-12	Office of the Comptroller of the Currency. <i>Sound Practices for Model Risk Management</i> . OCC 2011-12. April 4, 2011.
ECOA / Reg B	15 U.S.C. § 1691 et seq.; 12 C.F.R. Part 1002 (Regulation B).
EEOC 4/5ths Rule	29 C.F.R. § 1607 — Uniform Guidelines on Employee Selection Procedures.
CFPB Fair Lending	CFPB. <i>Supervisory Guidance on Model Risk Management</i> . 2021.